Topics

- some useful limits
- Series
- Comparison test
- Geometric series

Thm (3.20)

(a) If $p > 0$, $\lim_{n\to\infty} \frac{1}{n^p} = 0$

proof: Suppose $\varepsilon > 0$. $\frac{1}{n^p} < \varepsilon \iff \frac{1}{\varepsilon} < n^p \iff \left(\frac{1}{\varepsilon}\right)^{1/p} < n$

In particular, $n > \left(\frac{1}{\varepsilon}\right)^{1/p} \Rightarrow \frac{1}{n^p} < \varepsilon$.

(b) If $p > 0$, $\lim_{n\to\infty} \sqrt[n]{p} = 1$

proof: Let $p > 0$. want to show $\lim_{n\to\infty} \sqrt[n]{p} = 1$. Let $p > 1$.

put $x_n = \sqrt[n]{p} - 1$. want to show $x_n \to 0$

$$p = (x_n + 1)^n \geq 1 + n x_n$$

$$\frac{p-1}{n} \geq x_n$$

so $x_n \to 0$.

If $p = 1$, trivial.

If $p < 1$, $\sqrt[n]{1/p} \to 1 \Rightarrow \frac{1}{\sqrt[n]{p}} \to 1 \Rightarrow \sqrt[n]{p} \to 1$.

(c) $\lim_{n\to\infty} \sqrt[n]{n} = 1$

Proof: Let $x_n = \sqrt[n]{n} - 1$

$$(x_n+1)^n = n$$

$$1 + n x_n + \frac{n(n-1)}{2} x_n^2 \le n$$

$$x_n^2 \le \frac{2}{n-1} \qquad x_n \le \frac{\sqrt{2}}{\sqrt{n-1}} \to 0 \quad \text{by (a)}.$$

(d) If $|x| < 1$ then $\lim_{n\to\infty} x^n = 0$.

proof: It is equiv to show $|x^n| = |x|^n \to 0$, so we can assume $x \in (0,1)$.

$$x = \frac{1}{1+p} \quad \text{for some} \quad p \in (0,\infty).$$

$$x^n = \left(\frac{1}{1+p}\right)^n = \frac{1}{1 + np + \cdots + p^n} \le \frac{1}{np}$$

Since $\frac{1}{np} \to 0$, $x^n \to 0$

(e) If $|x|<1$, $\alpha \in \mathbb{Q}$ $\lim_{n\to\infty} x^n n^\alpha = 0$ (Exercize. Modify proof of (d))

Series

Def Given a sequence (a_n),

$$\sum_{k=p}^{q} a_k = a_p + \cdots + a_q$$

From a_n we get another sequence:

$$S_n = \sum_{k=1}^{n} a_k$$

(S_n) is called an <u>infinite series</u>, or a <u>series</u>. We denote (S_n) by $\sum a_n$.

Def If $\lim_{n\to\infty} S_n = S$, we say the series converges, and write

$$\sum_{n=1}^{\infty} a_n = S$$

If (S_n) diverges, we say $\sum_{n=1}^{\infty} a_n$ diverges.

Note: Sequences are the same thing as series. Any (s_n) is the series coming from $a_1 = s_1$, $a_k = s_k - s_{k-1}$ for $k > 1$.

Thm (3.22) $\sum a_n$ converges iff for $\forall \varepsilon > 0$, $\exists N$ s.t.

$$\left| \sum_{k=n}^{m} a_n \right| \leq \varepsilon$$

for all $m, n \in \mathbb{N}$ s.t. $N \leq n \leq m$.

Proof: This is just the Cauchy criterion:

$$\sum_{k=1}^{m} a_k - \sum_{k=1}^{n} a_k = \sum_{k=n+1}^{m} a_k$$

In particular:

Thm (3.23) If $\sum a_n$ converges, $\lim_{n\to\infty} a_n = 0$.

The condition $a_n \to 0$ is not sufficient to ensure $\sum a_n$ converges:

$\sum_{n=1}^{\infty} \frac{1}{n}$ diverges

Any thm about sequences is a thm about series:

Thm If $a_n \geq 0$, then $s_n = \sum_{k=1}^{n} a_k$ converges iff it is bounded.

Thm (3.25) [Comparison test]

(a) If $|a_n| \le c_n$ for $n \ge N_0$, and $\sum c_n$ converges, then $\sum a_n$ converges.

(b) If $a_n \ge d_n \ge 0$ for $n \ge N_0$, and $\sum d_n$ diverges, then $\sum a_n$ diverges. (Equiv: a_n converges $\Rightarrow$ d_n converges)

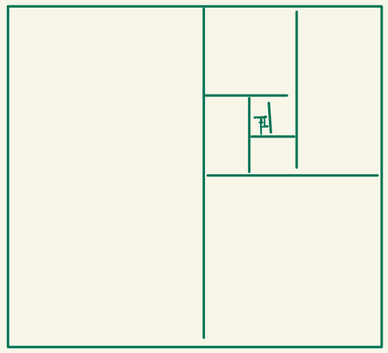
Proof: (a) Let $m \ge n$

$$\left| \sum_{k=n}^{m} a_k \right| \le \sum_{k=n}^{m} |a_k| \le \sum_{k=n}^{m} c_k$$

So Cauchy criterion for $\sum c_k$ implies Cauchy criterion for $\sum a_k$.

(b) This is special case of (a)

Geometric Series



$$1 = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots$$

Let's prove this.

Fundamental Identity

$$(1 + a + a^2 + \cdots + a^m)(1 - a) = (1 - a^{m+1})$$

$$\sum_{k=0}^{m} a^k = \frac{(1 - a^{m+1})}{1 - a}$$

Thm (3.26) If $0 \le x < 1$, then $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$.

If $x \ge 1$, series diverges.

Proof: If $x = 1$, $s_n = n$ which diverges.

If $x > 1$, $x^n \ge 1$, so Comparison test $\Rightarrow$ divergence.

Suppose $x \in [0,1)$.

$$s_n = \sum_{k=0}^{n} x^k = \frac{1-x^{n+1}}{1-x}$$

$$\lim_{n\to\infty} \frac{1-x^{n+1}}{1-x} = \frac{1-\lim_{n\to\infty} x^{n+1}}{1-x} = \frac{1}{1-x}$$